

Linear Algebra for Machine Learning: Practice Assignments

Assignment 1: Vector Operations Practice

Problem 1: Vector Addition and Scalar Multiplication

Given:

Vector **A** = [3, 5, 2]

Vector **B** = [1, 4, 6]

Scalar **k** = 2

Tasks:

1. Compute **A + B**
 2. Compute **A - B**
 3. Compute **k × A**
 4. Compute **(2A + 3B)**
 5. Find the magnitude (length) of **A** ($|A| = \sqrt{x_1^2 + x_2^2 + x_3^2}$)
-

Problem 2: Dot Products and Similarity

Given:

User Preference Vector **U** = [4, 2, 0, 5, 1]

Movie Genre Vectors:

M₁ = [1, 0, 1, 1, 0] (Action, Drama)

M₂ = [0, 1, 0, 0, 1] (Comedy, Romance)

M₃ = [1, 1, 0, 1, 1] (Mixed)

Tasks:

1. Calculate **U · M**₁
 2. Calculate **U · M**₂
 3. Calculate **U · M**₃
 4. Which movie is most similar to the user's preferences?
 5. What does a negative dot product imply about similarity?
-

Problem 3: Matrix Multiplication for Predictions

Given:

Feature Matrix **X** (3 students × 2 features):

Hours studied, Hours slept

text

[5, 8] ← Student 1

$X = [3, 7] \leftarrow \text{Student 2}$

$[7, 6] \leftarrow \text{Student 3}$

Weight Vector $\mathbf{w}$ (for predicting exam score):

text

$w = [2.5] \leftarrow \text{Weight for study hours}$

$[1.0] \leftarrow \text{Weight for sleep hours}$

Bias $\mathbf{b} = 50$

Tasks:

1. Compute $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w} + \mathbf{b}$ (predicted scores)
 2. Show step-by-step matrix multiplication
 3. If actual scores are: Student1=70, Student2=60, Student3=75, calculate errors
 4. What do the weights 2.5 and 1.0 represent in real terms?
-

Assignment 2: Linear Algebra in Simple Regression

Problem 4: Standardization

Given:

Original dataset: [12, 15, 18, 21, 24]

Mean (μ) = 18

Standard Deviation (σ) = 4.24

Tasks:

1. Standardize the first value: $z = (12 - 18) / 4.24$
 2. Standardize all values
 3. What is the mean of standardized values?
 4. What is the standard deviation of standardized values?
 5. Why do we standardize data in machine learning?
-

Problem 5: Simple Linear Regression Setup

Given data points:

(x, y) pairs: (1, 2), (2, 3), (3, 5), (4, 4)

Tasks:

1. Create feature matrix $\mathbf{X}$ with bias column
2. Create target vector $\mathbf{y}$

3. Write the Normal Equation: $\mathbf{w} = ?$
 4. Compute $\mathbf{X}^T \mathbf{X}$
 5. If $\mathbf{X}^T \mathbf{y} = [14, 34]^T$, find $\mathbf{w}$ (weights) using the Normal Equation
-

Problem 6: Understanding Matrix Operations

Given:

text

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \quad \begin{bmatrix} 5 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 4 \end{bmatrix} \quad x = \begin{bmatrix} 6 \end{bmatrix}$$

Tasks:

1. Compute A^T (transpose of A)
 2. Compute $A^T A$
 3. Compute Ax (matrix-vector multiplication)
 4. If $(A^T A)^{-1} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$, verify it's the inverse
 5. What does the inverse matrix allow us to do in linear regression?
-

Assignment 3: Geometric Interpretation

Problem 7: Dot Product Geometry

Given:

$$\mathbf{v}_1 = [3, 0] \text{ (points East)}$$

$$\mathbf{v}_2 = [0, 4] \text{ (points North)}$$

$$\mathbf{v}_3 = [2, 2] \text{ (points Northeast)}$$

$$\mathbf{v}_4 = [-3, 0] \text{ (points West)}$$

Tasks:

1. Calculate $\mathbf{v}_1 \cdot \mathbf{v}_2$
 2. Calculate $\mathbf{v}_1 \cdot \mathbf{v}_3$
 3. Calculate $\mathbf{v}_1 \cdot \mathbf{v}_4$
 4. What do these dot products tell you about the angles between vectors?
 5. Draw these vectors and verify your answers geometrically
-

Problem 8: Prediction Interpretation

Model equation:

$$\text{Score} = 10 + 2 \times \text{Study_Hours} + 1 \times \text{Sleep_Hours} - 0.5 \times \text{Phone_Hours}$$

Given student data:

Student A: [4, 8, 2] (Study, Sleep, Phone)

Student B: [6, 7, 4]

Student C: [2, 9, 6]

Tasks:

1. Predict scores for all students
 2. Which feature has the strongest positive impact?
 3. Which feature has negative impact?
 4. If Study_Hours increases by 1, how much does score change?
 5. What is the "base score" when all features are zero?
-

Problem 9: Error Calculation**Given predictions and actual values:**

Predicted: [65, 72, 80, 85]

Actual: [70, 70, 78, 88]

Tasks:

1. Calculate individual errors (predicted - actual)
 2. Calculate Mean Absolute Error (MAE)
 3. Calculate Mean Squared Error (MSE)
 4. Why do we square errors in MSE?
 5. Which error metric would be more sensitive to large errors?
-

Learning Objectives:

1. **Vector Operations:** Understand addition, subtraction, scalar multiplication
 2. **Dot Product:** Measure similarity and alignment between vectors
 3. **Matrix Operations:** Perform basic matrix multiplications
 4. **Standardization:** Prepare data for machine learning
 5. **Linear Regression Setup:** Formulate problems using matrices
 6. **Geometric Interpretation:** Visualize vector operations
 7. **Model Interpretation:** Understand weights and predictions
 8. **Error Metrics:** Evaluate model performance
-

Answer Key Framework

Problem 1:

1. [4, 9, 8]
2. [2, 1, -4]
3. [6, 10, 4]
4. [9, 22, 22]
5. $\sqrt{(3^2 + 5^2 + 2^2)} = \sqrt{38} \approx 6.16$

Problem 2:

1. $4 \times 1 + 2 \times 0 + 0 \times 1 + 5 \times 1 + 1 \times 0 = 9$
2. $4 \times 0 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 1 \times 1 = 3$
3. $4 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 1 + 1 \times 1 = 12$
4. Movie 3 (highest dot product = 12)
5. Negative dot product $\rightarrow$ vectors point in opposite directions $\rightarrow$ dissimilar preferences

Problem 3:

1. Predicted scores:
Student1: $5 \times 2.5 + 8 \times 1.0 + 50 = 70.5$
Student2: $3 \times 2.5 + 7 \times 1.0 + 50 = 64.5$
Student3: $7 \times 2.5 + 6 \times 1.0 + 50 = 73.5$
2. Matrix multiplication steps shown
3. Errors: [-0.5, -4.5, 1.5]
4. Each study hour adds 2.5 points, each sleep hour adds 1 point to base score of 50

Problem 4:

1. $(12-18)/4.24 \approx -1.41$
2. [-1.41, -0.71, 0, 0.71, 1.41]
3. Mean ≈ 0
4. Standard deviation ≈ 1
5. Standardization makes features comparable and helps optimization algorithms converge faster

Problem 5:

1. $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$
2. $y = [2, 3, 5, 4]^T$
3. $w = (X^T X)^{-1} X^T y$

4. $X^T X = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$
5. $w = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} \times \begin{bmatrix} 14 & 34 \end{bmatrix}^T = \begin{bmatrix} 6 & 1 \end{bmatrix}^T$

Problem 6:

1. $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
2. $A^T A = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$
3. $Ax = \begin{bmatrix} 17 & 39 \end{bmatrix}^T$
4. $(A^T A) \times (A^T A)^{-1} = I$ (identity matrix)
5. Inverse matrix allows solving linear equations for unknown weights

Problem 7:

1. $v_1 \cdot v_2 = 3 \times 0 + 0 \times 4 = 0$
2. $v_1 \cdot v_3 = 3 \times 2 + 0 \times 2 = 6$
3. $v_1 \cdot v_4 = 3 \times (-3) + 0 \times 0 = -9$
4. $v_1 \perp v_2$ (perpendicular), v_1 points same direction as v_3 , opposite to v_4
5. Drawing shows vectors on coordinate plane

Problem 8:

1. Student A: $10 + 2 \times 4 + 1 \times 8 - 0.5 \times 2 = 25$
 Student B: $10 + 2 \times 6 + 1 \times 7 - 0.5 \times 4 = 25$
 Student C: $10 + 2 \times 2 + 1 \times 9 - 0.5 \times 6 = 20$
2. Study_Hours (weight = 2)
3. Phone_Hours (weight = -0.5)
4. Score increases by 2 points
5. Base score = 10

Problem 9:

1. Errors: $[-5, 2, 2, -3]$
 2. $MAE = (5+2+2+3)/4 = 3$
 3. $MSE = (25+4+4+9)/4 = 10.5$
 4. Squaring emphasizes large errors and makes derivative continuous
 5. MSE is more sensitive to large errors
-